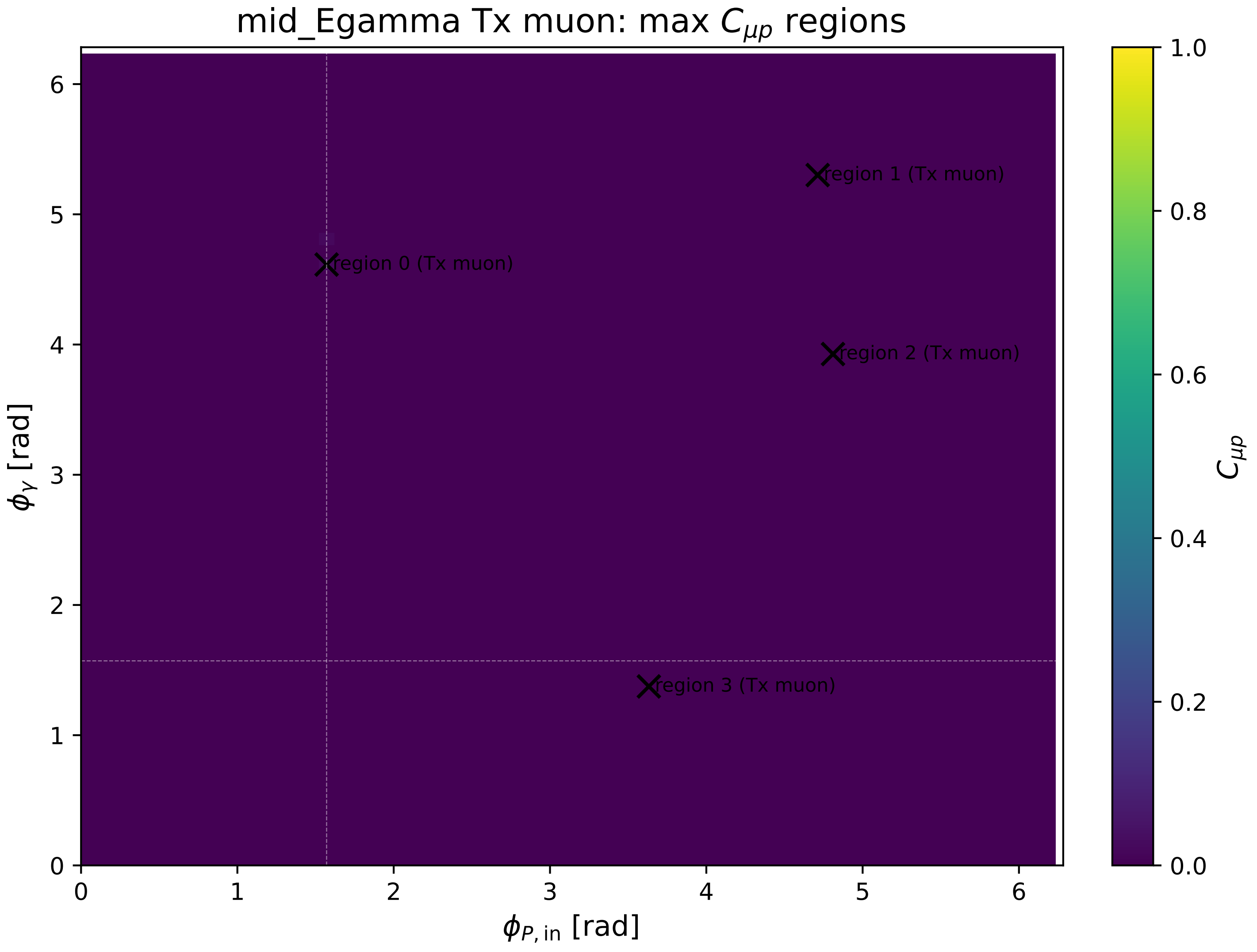
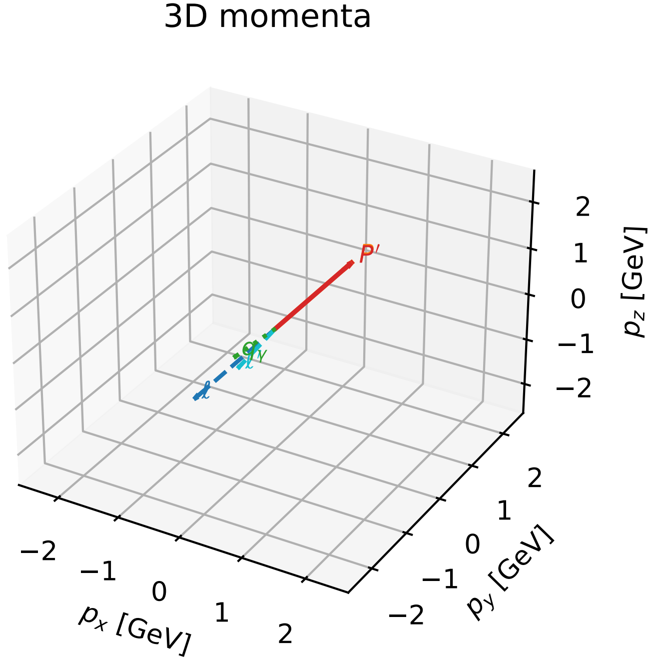


mid_Egamma Tx muon: max $C_{\mu p}$ regions
region 1 (Tx muon)
region 0 (Tx muon)
region 2 (Tx muon)
region 3 (Tx muon)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\mu p}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Tx muon characteristic max $C_{\mu p}$ configuration

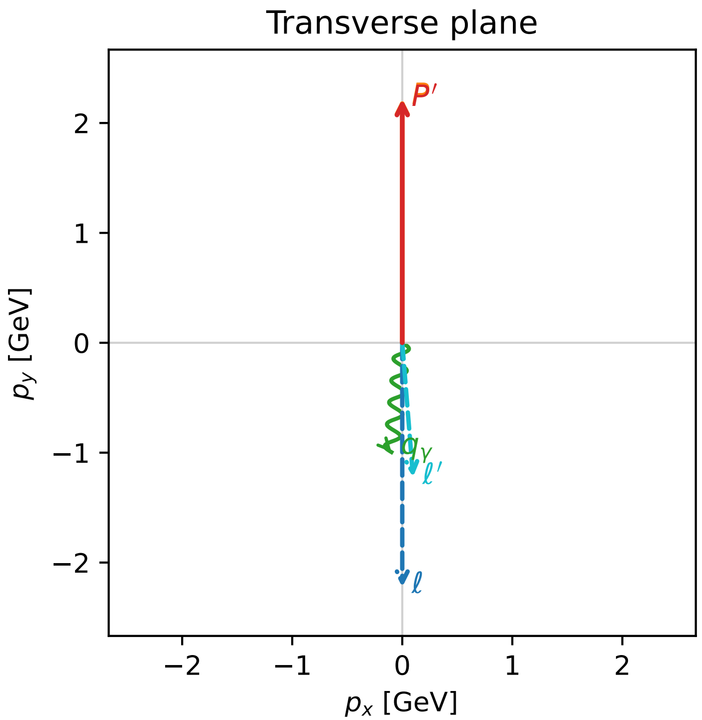


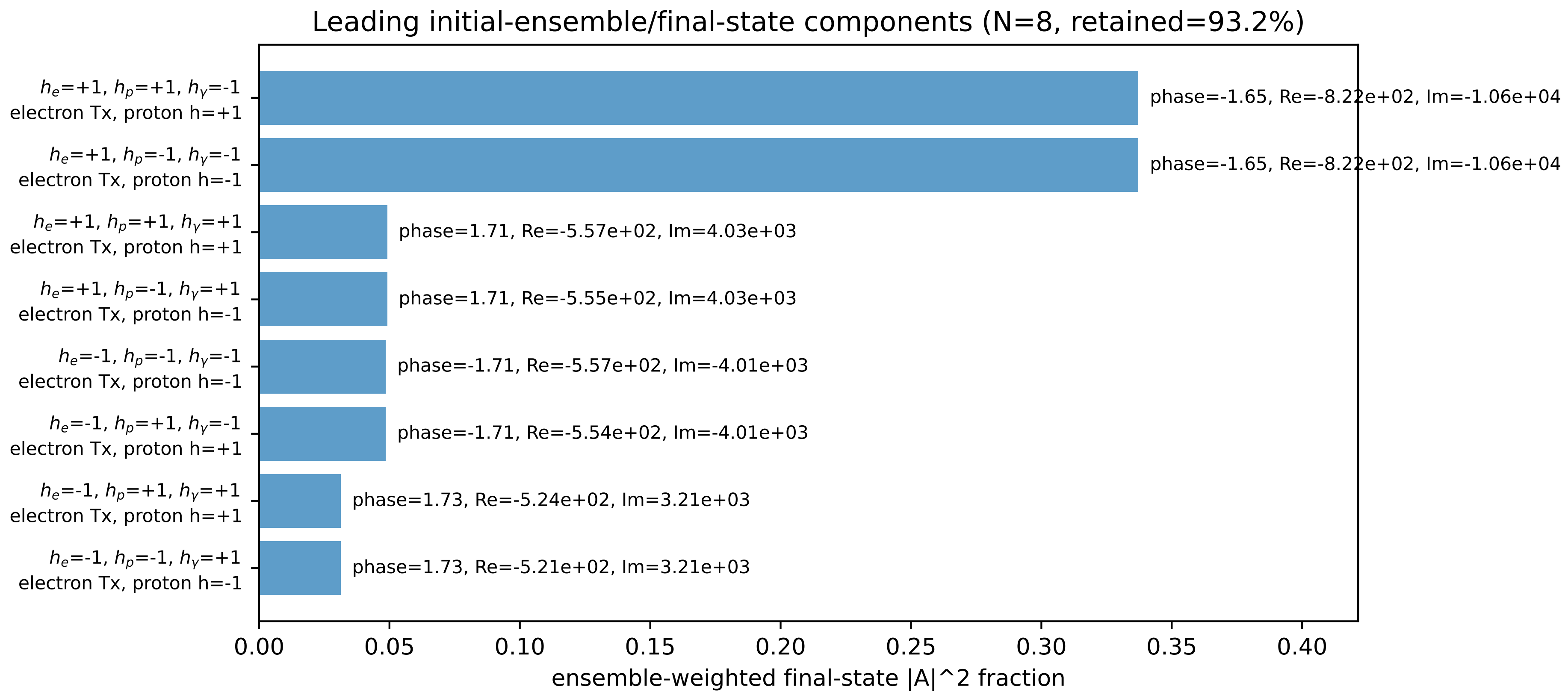

```
c_mu_p_mid_egamma_tx_lepton_region_0  C_{mup}=0.0219239
region: mid_Egamma_Tx_lepton_region_0  final-pair delta_xy=3.06157 rad
outgoing-state purity: 0.503601
kinematic point: high_s_high_theta_in_mid_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{x,\mu}, h_p)$

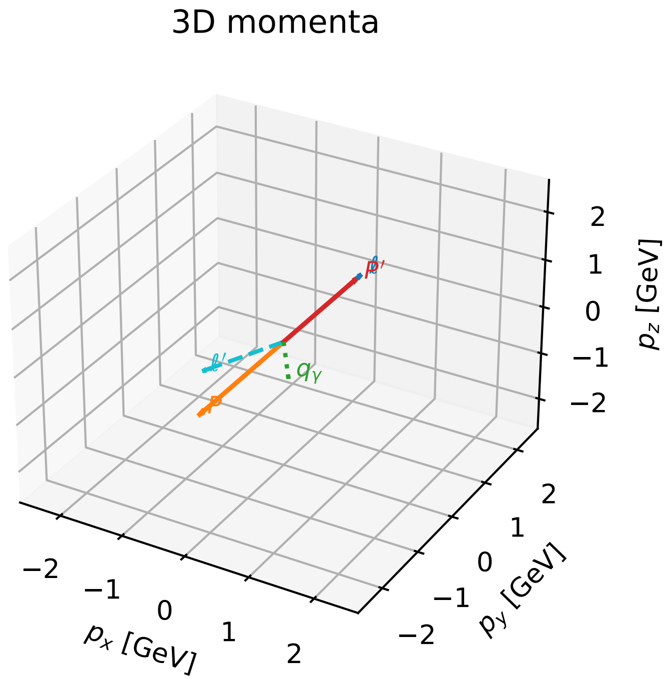
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.2175
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=1, ϕ_γ=4.61421
Q^2=0.021497, x_B=0.00232384, t=-4.76223e-06, W^2=10.109, y=0.448518
$\theta(\ell', \gamma)$=10.2097 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -0.0000 p_y= -2.2231 p_z= -0.0000
P E= 2.4129 p_x= 0.0000 p_y= 2.2231 p_z= 0.0000
ℓ' E= 1.2308 p_x= 0.0980 p_y= -1.2223 p_z= 0.0000
P' E= 2.4077 p_x= 0.0000 p_y= 2.2175 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.0980 p_y= -0.9952 p_z= 0.0000





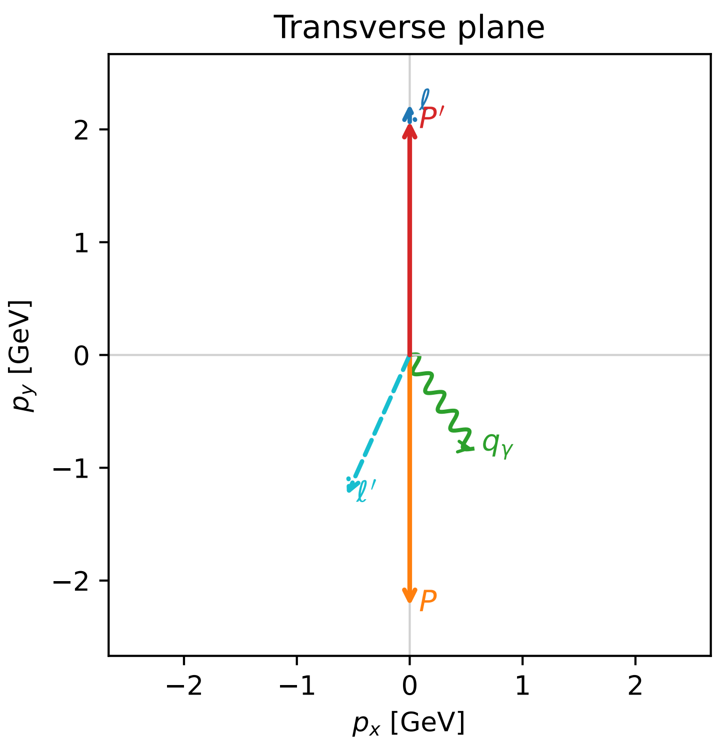
Tx muon characteristic max $C_{\mu p}$ configuration

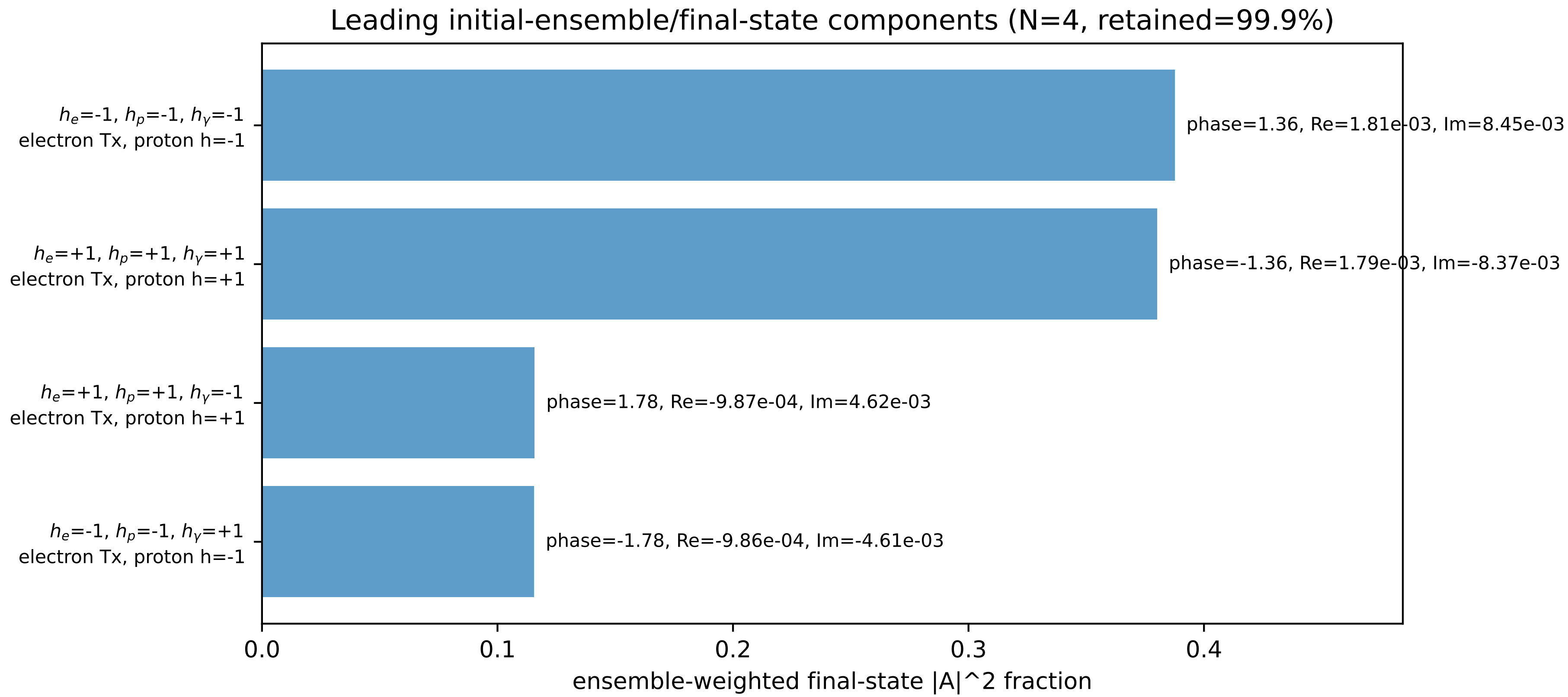


c_mu_p_mid_egamma_tx_lepton_region_1 C_{mup}=0.00141382
region: mid_Egamma_Tx_lepton_region_1 final-pair delta_xy=2.72065 rad
outgoing-state purity: 0.50002
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{x,\mu}, h_p)$

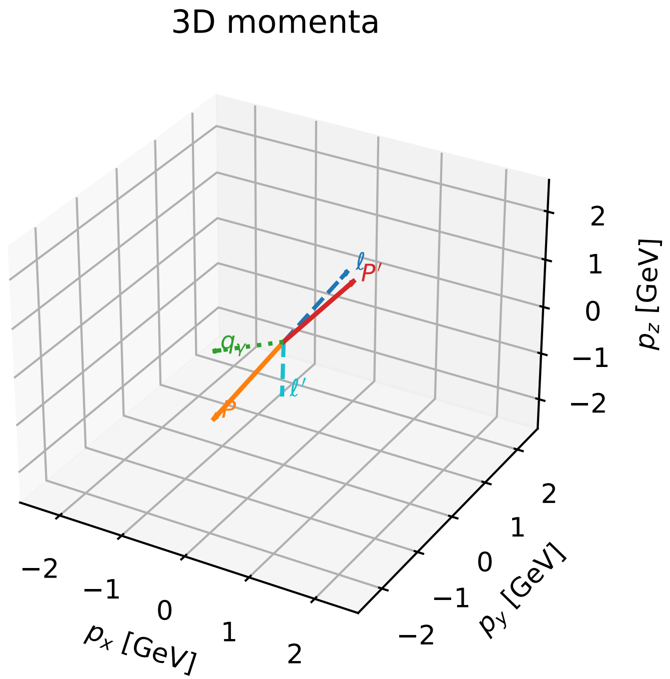
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.0724
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=5.30144
Q^2=11.5654, x_B=0.59124, t=-18.4324, W^2=8.87572, y=0.948432
$\theta(\ell', \gamma)$=57.8682 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 0.0000 p_y= 2.2231 p_z= -0.0000
P E= 2.4129 p_x= -0.0000 p_y= -2.2231 p_z= 0.0000
ℓ' E= 1.3637 p_x= -0.5556 p_y= -1.2409 p_z= 0.0000
P' E= 2.2748 p_x= 0.0000 p_y= 2.0724 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.5556 p_y= -0.8315 p_z= 0.0000





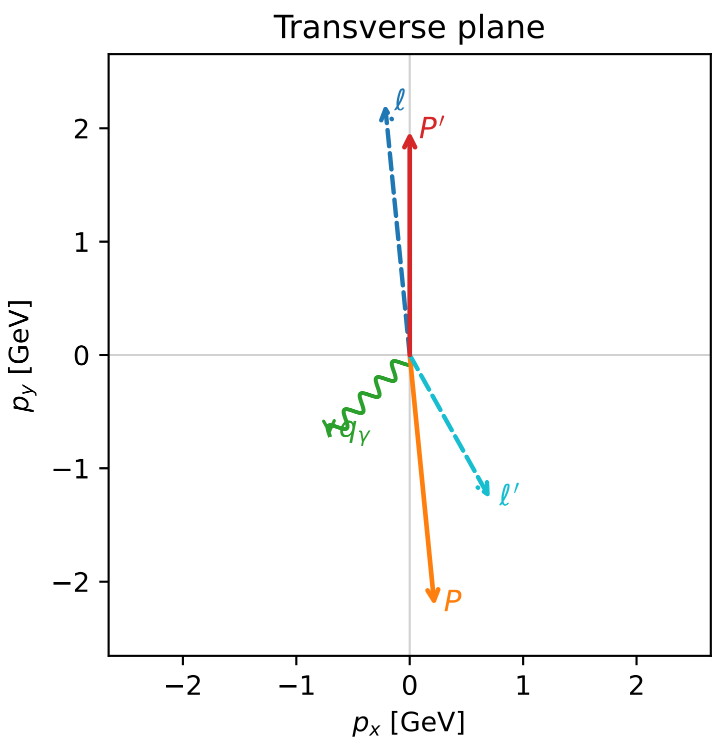
Tx muon characteristic max $C_{\mu p}$ configuration

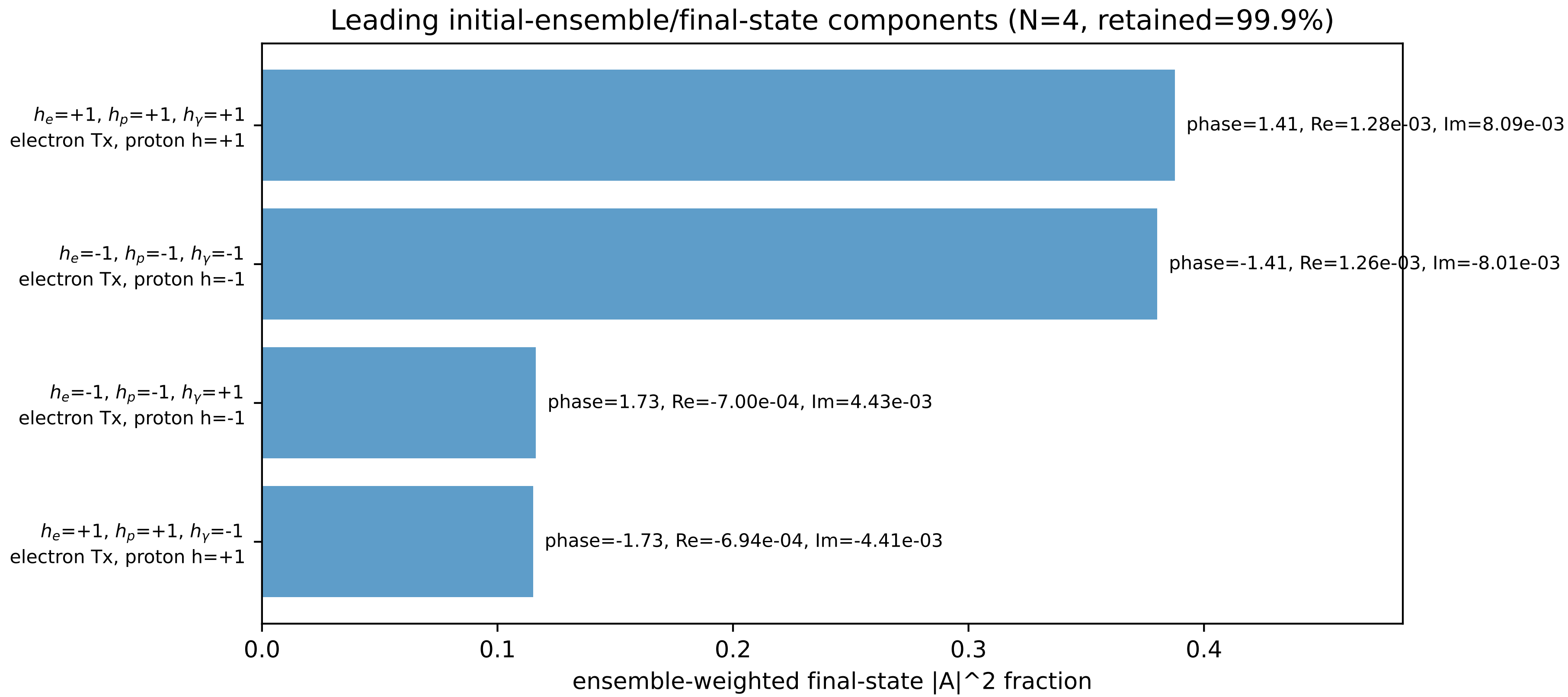


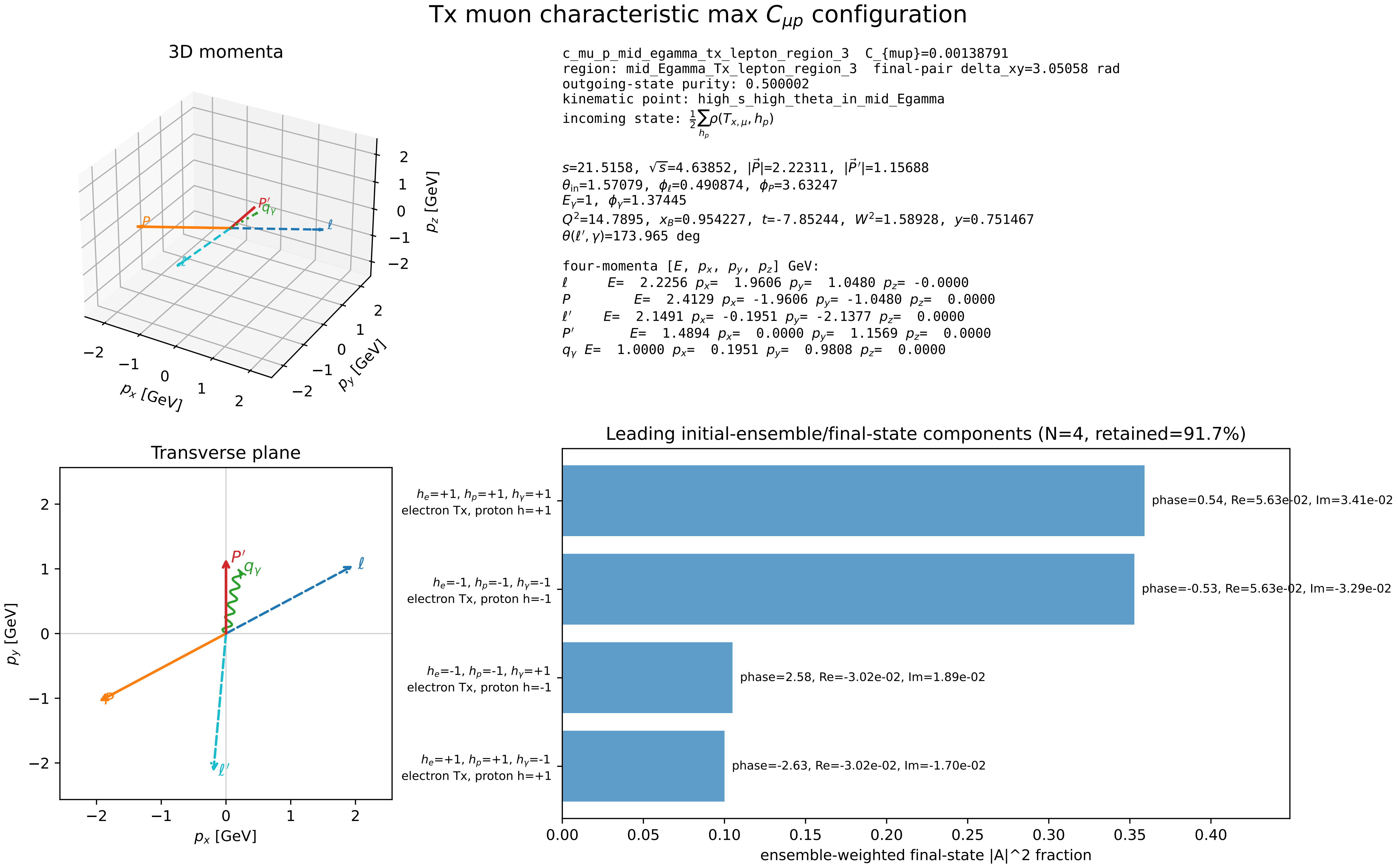
c_mu_p_mid_egamma_tx_lepton_region_2 C_{mup}=0.00140329
region: mid_Egamma_Tx_lepton_region_2 final-pair delta_xy=2.63219 rad
outgoing-state purity: 0.500016
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{x,\mu}, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.97303
θ_{in}=1.57079, ϕ_ℓ=1.66897, ϕ_P=4.81056
E_γ=1, ϕ_γ=3.92699
Q^2=12.3588, x_B=0.633188, t=-17.5133, W^2=8.03943, y=0.946355
$\theta(\ell', \gamma)$=74.1864 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -0.2179 p_y= 2.2124 p_z= -0.0000
P E= 2.4129 p_x= 0.2179 p_y= -2.2124 p_z= 0.0000
ℓ' E= 1.4539 p_x= 0.7071 p_y= -1.2659 p_z= 0.0000
P' E= 2.1847 p_x= 0.0000 p_y= 1.9730 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.7071 p_y= -0.7071 p_z= 0.0000






Tx muon characteristic max $C_{\mu p}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
P'
q_γ
ℓ
ℓ'
c_mu_p_mid_egamma_tx_lepton_region_3 C_{mup}=0.00138791
region: mid_Egamma_Tx_lepton_region_3 final-pair delta_xy=3.05058 rad
outgoing-state purity: 0.500002
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{x,\mu}, h_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.15688
θ_{in}=1.57079, ϕ_ℓ=0.490874, ϕ_P=3.63247
E_γ=1, ϕ_γ=1.37445
Q^2=14.7895, x_B=0.954227, t=-7.85244, W^2=1.58928, y=0.751467
$\theta(\ell', \gamma)$=173.965 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 1.9606 p_y= 1.0480 p_z= -0.0000
P E= 2.4129 p_x= -1.9606 p_y= -1.0480 p_z= 0.0000
ℓ' E= 2.1491 p_x= -0.1951 p_y= -2.1377 p_z= 0.0000
P' E= 1.4894 p_x= 0.0000 p_y= 1.1569 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.1951 p_y= 0.9808 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=4, retained=91.7%)
h_e=+1, h_p=+1, h_γ=+1
electron Tx, proton h=+1
phase=0.54, Re=5.63e-02, Im=3.41e-02
h_e=-1, h_p=-1, h_γ=-1
electron Tx, proton h=-1
phase=-0.53, Re=5.63e-02, Im=-3.29e-02
h_e=-1, h_p=-1, h_γ=+1
electron Tx, proton h=-1
phase=2.58, Re=-3.02e-02, Im=1.89e-02
h_e=+1, h_p=+1, h_γ=-1
electron Tx, proton h=+1
phase=-2.63, Re=-3.02e-02, Im=-1.70e-02
ensemble-weighted final-state |A|^2 fraction